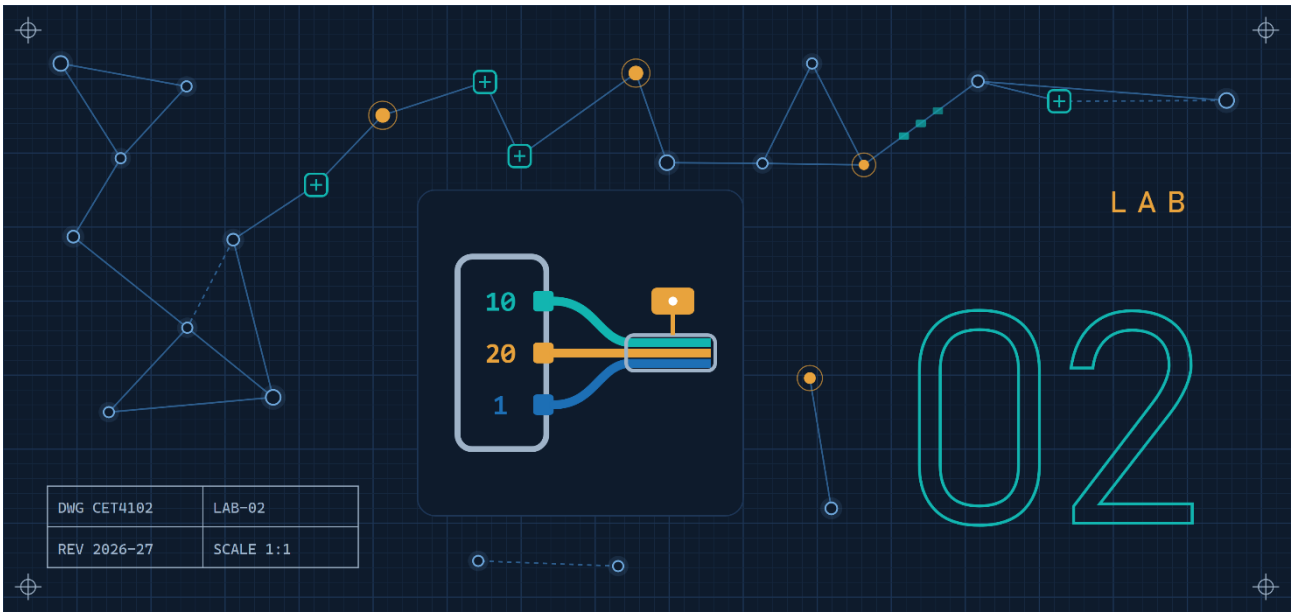
THE ISLAMIC UNIVERSITY
College of Technical Engineering
Department of Computer Techniques Engineering



LAB 02 · WEEK 2

Virtual LANs (VLANs)

Split one switched network into separate broadcast domains: create and name VLANs, assign access ports, verify them and test which hosts can still reach each other.

Broadcast domains and why VLANs are used	Creating and naming VLANs	Access ports and interface range
Verification with show vlan brief	Ping tests within and between VLANs	Preview: an 802.1Q trunk

Course	Computer Networks Protocols	**Code**	CET4102 · 6 ECTS
Stage	Fourth stage · Semester 7	**Session**	Laboratory session · Week 2
Instructor	Asst. Lect. Laith H. Alzubaidi	**Academic year**	2026 – 2027

Prepared for the fourth-stage students of the Department of Computer Techniques Engineering. Primary text: J. F. Kurose & K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Recommended: D. E. Comer, *Internetworking with TCP/IP*.

LEARNING OUTCOMES

By the end of this lab you will be able to:

1. explain why one switched LAN is one broadcast domain and how VLANs split it;
2. create and name VLANs on a Cisco 2960 switch and assign access ports to them;
3. check VLAN membership with `show vlan brief`;
4. predict and test which pings work inside one VLAN and between VLANs;
5. join two switches with an 802.1Q trunk and describe the tag it adds to each frame.

L2.1 Background

A switch learns which MAC address is behind each port and sends a frame for a known MAC address only out of that port (Lecture 5). A **broadcast** frame, such as an ARP request to FFFF.FFFF.FFFF, is different: the switch copies it out of every other port. The hosts that receive each other's broadcasts form a **broadcast domain**. By default, all ports of all connected switches form one broadcast domain.

In a large LAN this becomes a problem. Every host must process every broadcast, one faulty host can flood everybody, and all user groups (staff, students, guests) share one network. A router stops broadcasts, but a router port and a switch for every group is expensive.

KEY DEFINITION

A **virtual LAN (VLAN)** is a group of switch ports that the switch treats as a separate LAN, and therefore as a separate broadcast domain, although it shares the same switches and cables with other VLANs. Membership is set by configuration, not by wiring. The standard is IEEE 802.1Q.

The figure below shows one broadcast before and after the ports of a switch are split into two VLANs.

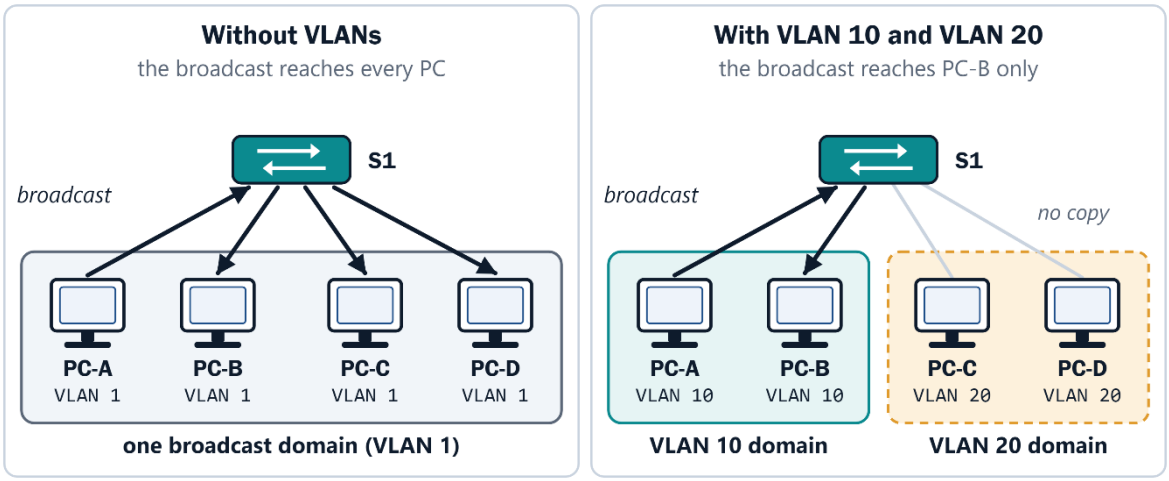


Figure L2.1 One switch and four PCs (a general example, not the lab topology). Left: all ports are in VLAN 1, so a broadcast from PC-A is copied to every other PC. Right: the ports are split into VLAN 10 and VLAN 20, so the same broadcast reaches only PC-B.

Three rules follow, and this lab tests each of them:

- **One VLAN = one IP subnet.** Here VLAN 10 uses 192.168.10.0/24 and VLAN 20 uses 192.168.20.0/24.

- **A switch never forwards a frame from one VLAN into another.** Traffic between VLANs must be routed by a router or a Layer-3 switch (Lab 3).
- **A VLAN can span several switches**, but the link between them must carry that VLAN.

A port that belongs to one VLAN is an **access port**. It connects an end device, and its frames are ordinary Ethernet frames: the PC does not know its VLAN. A link that carries several VLANs, usually between two switches or between a switch and a router (Lab 3), is a **trunk**. On a trunk the switch adds a small tag with the VLAN number to each frame (Step 7).

L2.2 Topology and addressing

Two Catalyst 2960-24TT switches, S1 and S2, have three PCs each. VLAN 10 (STAFF) and VLAN 20 (STUDENTS) exist on both switches. The third number of every subnet equals the VLAN ID, so an address shows its VLAN at a glance. The port plan is the same on both switches: Fa0/1–Fa0/10 in VLAN 10, Fa0/11–Fa0/20 in VLAN 20, and Fa0/21–Fa0/24 and G0/2 unused in VLAN 1. G0/1 joins the two switches.

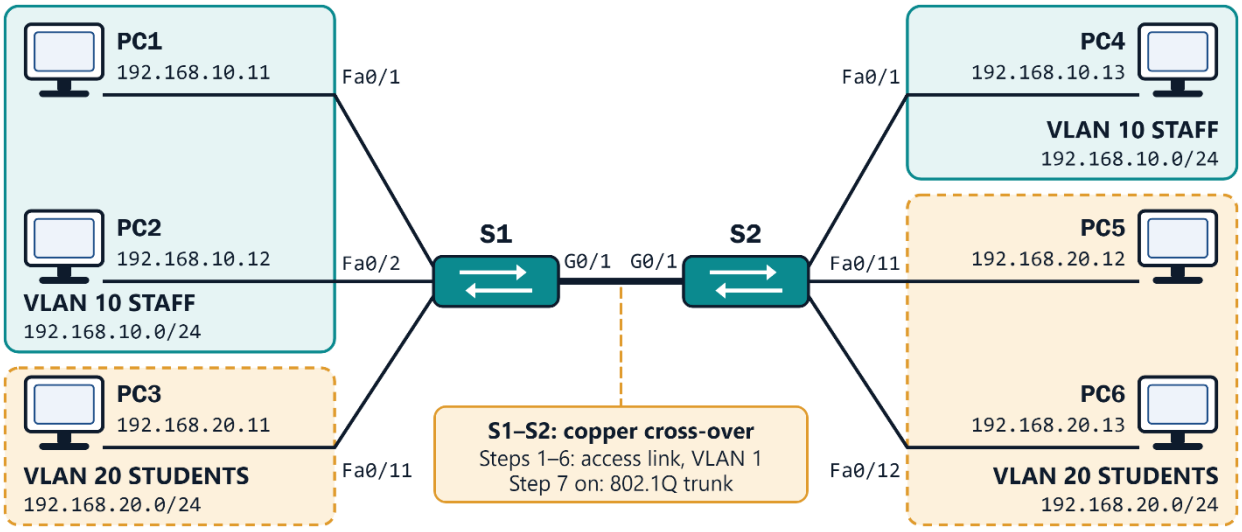


Figure L2.2 Lab topology. PCs use copper straight-through cables; S1 G0/1 and S2 G0/1 are joined by a copper cross-over cable, which becomes an 802.1Q trunk in Step 7.

Table L2.1 Addressing table. The gateways (.1) are added in Lab 3 (inter-VLAN routing).

Device	Interface	IP address	Subnet mask	Default gateway
PC1	Fa0 (to S1 Fa0/1)	192.168.10.11	255.255.255.0	192.168.10.1
PC2	Fa0 (to S1 Fa0/2)	192.168.10.12	255.255.255.0	192.168.10.1
PC3	Fa0 (to S1 Fa0/11)	192.168.20.11	255.255.255.0	192.168.20.1
PC4	Fa0 (to S2 Fa0/1)	192.168.10.13	255.255.255.0	192.168.10.1
PC5	Fa0 (to S2 Fa0/11)	192.168.20.12	255.255.255.0	192.168.20.1
PC6	Fa0 (to S2 Fa0/12)	192.168.20.13	255.255.255.0	192.168.20.1

BEFORE YOU START

- Revise switch learning, flooding and ARP (Lecture 5) and the IOS command modes (Lab 1).
- Copy the ping table from the Verification section and write your *predicted* result for every pair.
- Answer in one line: how many broadcast domains does the topology have before any VLAN is configured?

L2.3 Procedure

L2.3.1 Step 1: Build the network and configure the PCs

1. Place two **2960-24TT** switches and six **PC-PT** hosts, and name them S1, S2 and PC1–PC6.
2. Cable them as in the topology figure. Wait until every link light is green: a new switch port stays amber for about 30 s (**Fast Forward Time** skips the wait).
3. On each PC open **Desktop > IP Configuration**, select **Static** and enter the values from the addressing table.
4. Give each switch its basic settings (S2 is the same with `hostname S2`):

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
S1(config)# no ip domain-lookup
S1(config)# line console 0
S1(config-line)# logging synchronous
S1(config-line)# end
```

L2.3.2 Step 2: Observe the default state

Out of the box every port is in VLAN 1. Check it with `show vlan brief` on S1: all 24 Fast Ethernet ports and both Gigabit ports, including G0/1 towards S2, are listed under `1 default`. So the two switches form one broadcast domain. The list ends with VLANs 1002–1005, which are reserved for old FDDI and Token Ring networks. (Packet Tracer prints `Gig0/1` and status `active`; a real Catalyst prints `Gi0/1` and `act/unsup` for 1002–1005.)

Next, run the six pings of the ping table (Verification section) from the PCs (**Desktop > Command Prompt**). Record the results in the "Before VLANs" column. Rows 1, 2, 4 and 5 work, because all PCs share VLAN 1. Rows 3 and 6 fail for a **Layer-3** reason: the two PCs are in different subnets. So PC1 tries to send the packet to its gateway 192.168.10.1, which does not exist yet.

Now watch a broadcast. In **Simulation** mode, use **Edit Filters** to show only ARP and ICMP events. On PC1 type `arp -d` (empty the ARP cache) and ping PC4 again. Step with **Capture/Forward**: the ARP request is copied to **all five** other PCs. Take a screenshot, then return to Realtime mode.

L2.3.3 Step 3: Create and name the VLANs

Each VLAN has a number, the **VLAN ID** (1–4094), and an optional name. VLAN 1 is the **default VLAN**: every port starts in it, and it cannot be deleted. IDs 1–1005 form the **normal range**. VLAN 1 and VLANs 1002–1005 already exist, so your own VLANs take IDs 2–1001. (IDs 1006–4094, the extended range, are not used here.)

`vlan 10` creates VLAN 10, and `name` labels it. The name is only a label for people; switches identify a VLAN by its ID. Enter on S1, then on S2:

```
S1# configure terminal
S1(config)# vlan 10
S1(config-vlan)# name STAFF
S1(config-vlan)# exit
S1(config)# vlan 20
S1(config-vlan)# name STUDENTS
S1(config-vlan)# end
```

Run `show vlan brief` again: VLANs 10 STAFF and 20 STUDENTS are listed as **active** but have **no ports** yet, so nothing has changed for the PCs. IOS saves these VLANs at once in the file `flash:vlan.dat`, not in the running-config.

L2.3.4 Step 4: Assign the access ports

Two interface commands make a port an access port of a VLAN. `switchport mode access` fixes the port as an access port, and `switchport access vlan 10` puts it in VLAN 10. `interface range` applies them to many ports at once. Enter the same commands on S1 and on S2:

```
S1# configure terminal
S1(config)# interface range fastEthernet 0/1 - 10
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 10
S1(config-if-range)# exit
S1(config)# interface range fastEthernet 0/11 - 20
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 20
S1(config-if-range)# end
S1# copy running-config startup-config
```

Unlike the VLANs, these port commands are kept in the running-config, so the last line saves them. For one port use, for example, `interface fastEthernet 0/2` (prompt `S1(config-if)#`). `no switchport access vlan` returns a port to VLAN 1.

> **COMMON MISTAKE**
>
> `switchport mode access vlan 10` is not a valid command, although it appears in some lab sheets: the mode and the VLAN are two separate commands. Also, if you give a VLAN that does not exist, IOS creates it (`% Access VLAN does not exist. Creating vlan 30`). Create and name your VLANs first.

L2.3.5 Step 5: Verify VLAN membership

```
S1# show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
10   STAFF                            active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10
20   STUDENTS                         active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
```

```
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Compare every line with the port plan (Topology and addressing). A port is listed even without a cable (Fa0/3–Fa0/10 have no PC). G0/1 is still in VLAN 1.

L2.3.6 Step 6: Test pings within and between VLANs

Repeat the pings of the ping table and fill the "After VLANs" column. You will see three cases:

- **PC1 → PC2** (same switch, same VLAN) works.
- **PC1 → PC3** (same switch, different VLANs) fails: the switch never passes a frame from VLAN 10 to VLAN 20.
- **PC1 → PC4** (same VLAN, different switches) now fails: both ends of the S1–S2 link are access ports in VLAN 1, so the link cannot carry VLAN 10 frames.

In Simulation mode, run `arp -d` on PC1 again and ping PC2. The ARP request now reaches **only PC2**: the broadcast stays inside VLAN 10. Take a second screenshot.

L2.3.7 Step 7: Join the switches with an 802.1Q trunk (preview)

Why is G0/1 not a trunk already? A 2960 port starts in *dynamic auto* mode: it accepts a trunk when the other end asks, but never asks itself, so two auto ports stay access ports. Set trunk mode by hand on S1, then on S2 (trunk options follow in Lab 3):

```
S1# configure terminal
S1(config)# interface gigabitEthernet 0/1
S1(config-if)# switchport mode trunk
S1(config-if)# end
S1# show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      1-1005

Port        Vlans allowed and active in management domain
Gig0/1      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      1,10,20
```

G0/1 is now an 802.1Q trunk (mode **on**) carrying the active VLANs 1, 10 and 20 (a real switch shows allowed VLANs 1-4094). Frames of the **native VLAN** (VLAN 1) cross untagged (Lab 3). Gig0/1 also leaves `show vlan brief`, which lists access ports only.

How does S2 know the VLAN of each frame? Before sending a frame on the trunk, S1 inserts a 4-byte **802.1Q tag** (figure below).

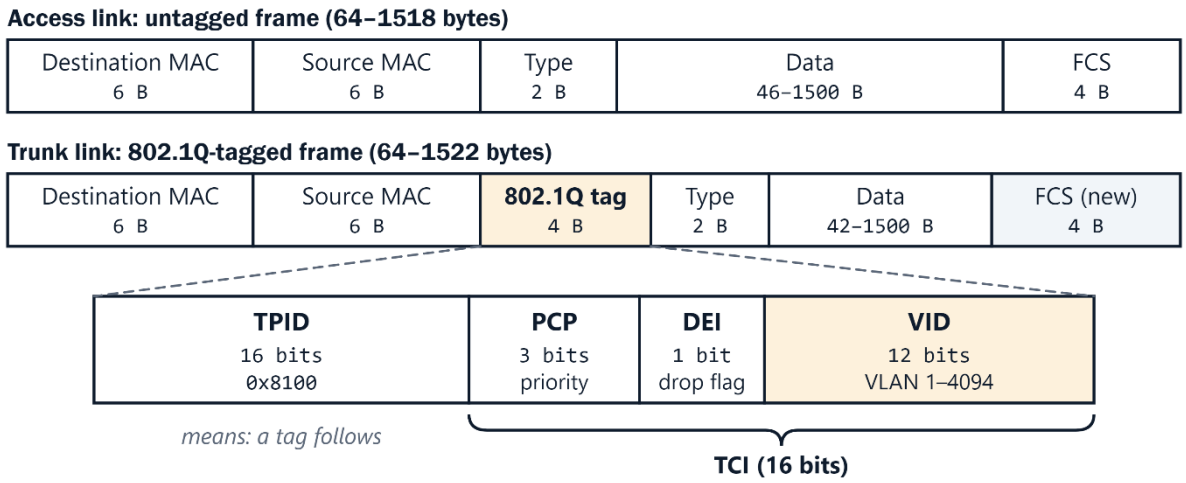


Figure L2.3 An access link carries ordinary frames. On a trunk the switch inserts a 4-byte 802.1Q tag after the source MAC address and recomputes the FCS; the 12-bit VID field names the VLAN.

The **TPID** (16 bits, always 0x8100) says that a tag follows. The **TCI** (16 bits) holds a 3-bit priority (PCP), a 1-bit drop-eligible flag (DEI) and the 12-bit **VLAN ID** (VID). Twelve bits give 4096 values; 0 and 4095 are reserved, so VLANs 1–4094 can be used. The receiving switch reads the VID, removes the tag and forwards the frame only to ports of that VLAN.

> **WORKED EXAMPLE: THE TAG OF A VLAN 20 FRAME**
>
> ```
> PCP = 000    DEI = 0    VID = 20 = 0000 0001 0100
> TCI = 000 0 0000 0001 0100 = 0x0014
> tag = TPID + TCI = 81 00 00 14
> ```
>
> S2 reads VID 0x014 = 20 and sends the frame only out of its VLAN 20 ports.

Fill the "After trunk" column. PC1 → PC4 and PC3 → PC5 now work; PC1 → PC3 and PC1 → PC5 still fail, because a trunk carries many VLANs but does not route between them. In Simulation mode, ping PC1 → PC4 again and click the frame on the S1–S2 link. Its details show an 802.1Q (Dot1q) header: TPID 0x8100 and VLAN ID 10.

L2.4 Verification

On both switches, `show vlan brief` must list VLANs 10 and 20 with the ports of the plan (Gig0/1 not listed), and `show interfaces trunk` must show Gig0/1 trunking with VLANs 1,10,20 active. The PC pings must match the table below. Save both switches and keep the .pkt file: Lab 3 continues from it.

Table L2.2 Ping table: expected results.

#	Pair	Relation	Before VLANs	After VLANs	After trunk
1	PC1 → PC2	same switch, same VLAN	works	works	works
2	PC5 → PC6	same switch, same VLAN	works	works	works
3	PC1 → PC3	same switch, different VLANs	fails	fails	fails
4	PC1 → PC4	different switches, same VLAN	works	**fails**	works
5	PC3 → PC5	different switches, same VLAN	works	**fails**	works
6	PC1 → PC5	different switches, different VLANs	fails	fails	fails

Rows 3 and 6 fail in every column. Before Step 4 the reason is Layer 3: different subnets and no router. After Step 4 they also fail at Layer 2: different VLANs. Rows 4 and 5 show what the trunk adds.

L2.5 Common problems

Table L2.3 Common problems and their fixes.

Symptom	Likely cause	Fix
PC2 cannot ping PC1	Fa0/2 left in VLAN 1 (see `show vlan brief`)	`switchport access vlan 10` on Fa0/2
Same-VLAN pings between switches fail; same-switch pings work	G0/1 is not a trunk (`show interfaces trunk` shows nothing)	`switchport mode trunk` on G0/1 of both switches
PC5 and PC6 reach nobody; their ports vanished from `show vlan brief`	VLAN 20 was deleted with `no vlan 20`; its ports became inactive	`vlan 20`, `name STUDENTS`; the ports return by themselves
One PC reaches nobody	its IP subnet does not match the VLAN of its port	correct the PC address or the port VLAN

L2.6 Discussion questions

1. What is a VLAN, and which problem of a flat switched LAN does it solve?
2. Why did PC1 → PC4 fail after Step 4 but work after Step 7?
3. Compare an access port and a trunk port: number of VLANs, tagging, typical use and configuration command.
4. Write the 802.1Q tag bytes for a frame of VLAN 30 with priority 0. Then decode the tag `81 00 A0 14`.
5. You erased the startup-config of S1 and reloaded it, but `show vlan brief` still lists VLANs 10 and 20. Explain why.
6. VLAN 10 is named STAFF on S1 but LECTURERS on S2. Does PC1 → PC4 still work? What must match on the two switches?

L2.7 Lab report

- The ping table with your predicted and observed results, and a one-line reason for every failure.
- From S1: `show vlan brief` (after Steps 5 and 7) and `show interfaces trunk`; the two Simulation-mode screenshots of Steps 2 and 6.
- Your answers to the discussion questions and your final .pkt file.